

Buck Converter Design Documentation

Table of Contents

1. Introduction
2. Project Overview
3. Design Iterations
4. Component Selection and Calculations
5. Component List
6. Conclusion

1. Introduction

This document details the development and implementation of a Buck converter using the XL2013E1 IC. The converter is designed to step down 12V DC input to 5V DC output for industrial applications.

2. Project Overview

- Input Voltage: 12V DC
- Output Voltage: 5V DC
- Switching Frequency: 150 Hz
- Controller IC: XL2013E1

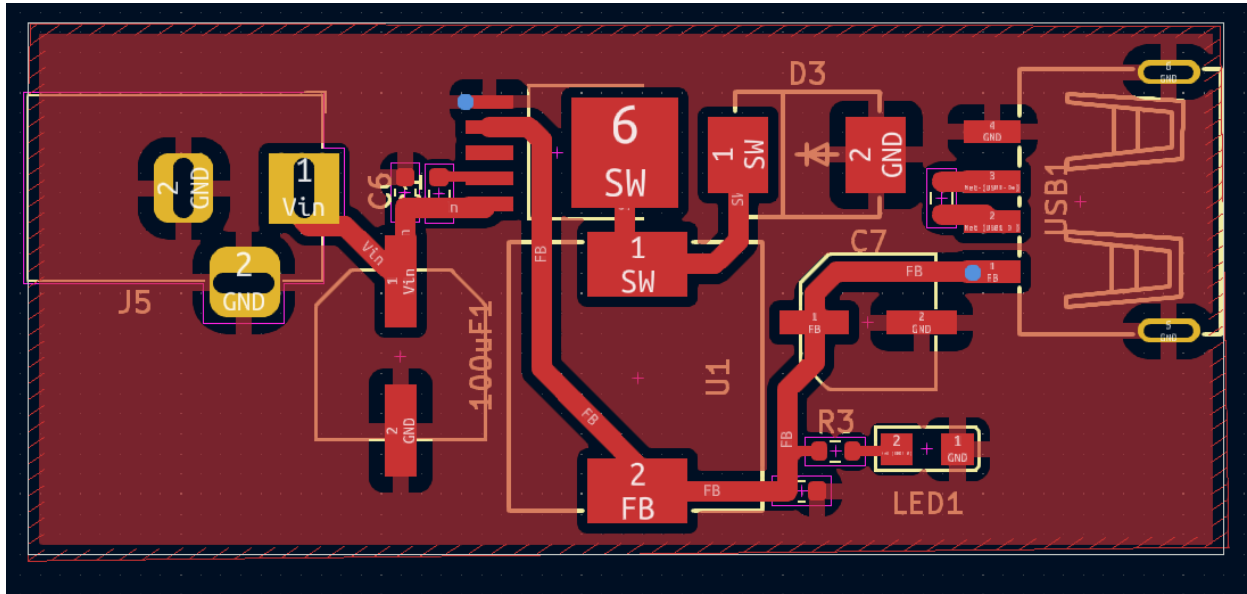
3. Design Iterations

3.1 First Iteration

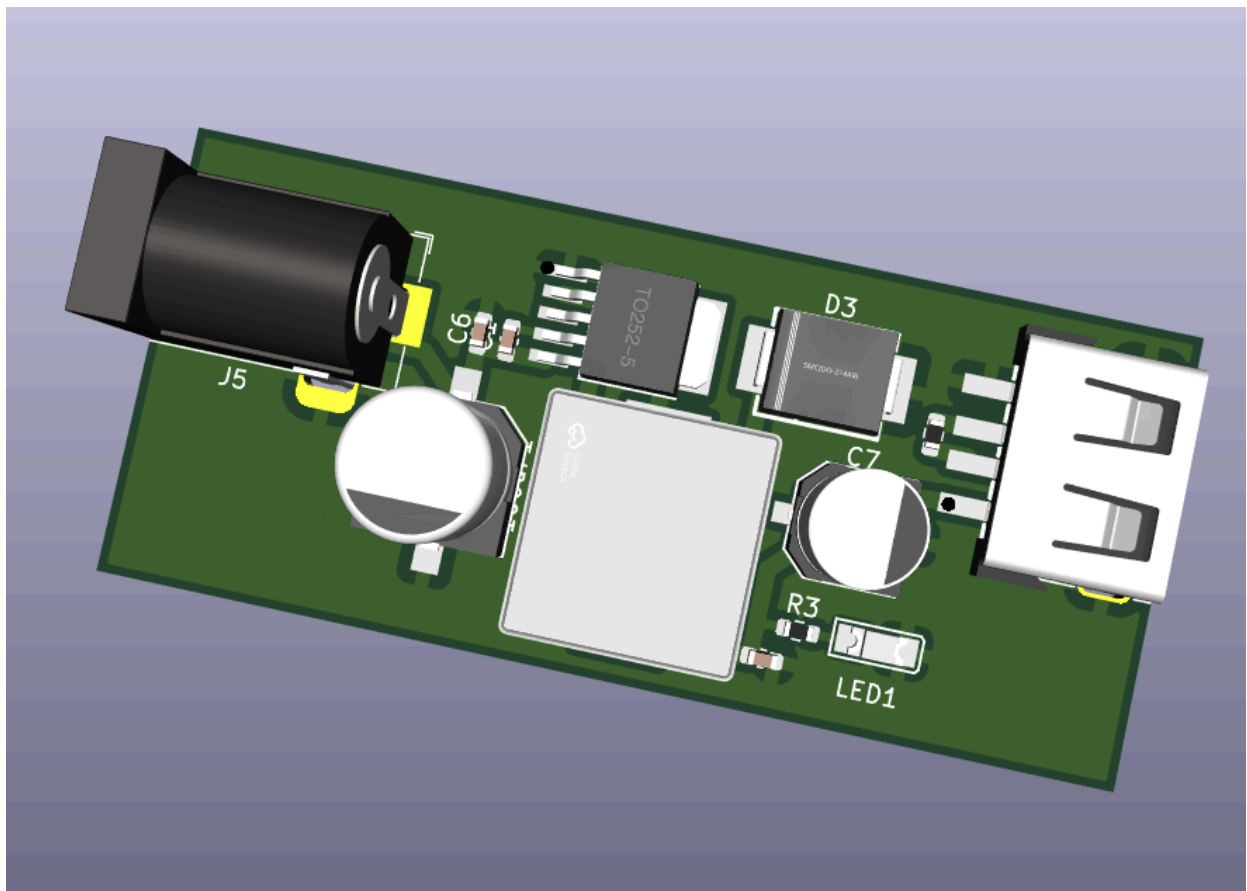
The first iteration focused on basic functionality validation and learning:

Design Approach

- Basic implementation using XL2013E1 controller
- Component selection based on general guidelines without detailed calculations
- Simple PCB layout using KiCAD
- Board size: 60mm × 26mm



PCB Layout



Component Selection

- Inductor: 33 μ H (selected from available components)
- Input Capacitor: 330 μ F electrolytic
- Output Capacitor: 330 μ F electrolytic

Observations and Calculations:

Source Voltage(V)	Source current(A)	Average Voltage(V)	Range Fluctuation(mV)	Current Drawn(I)	Remarks
10	0.43	4.58	700-960	0.85	20E
11	0.39	4.52	720-800	0.84	20E
12	0.35	4.54	520-800	0.84	20E
13	0.33	4.54	600-800	0.84	20E
14	0.30	4.56	580-840	0.84	20E
15	0.29	4.64	380-520	0.84	20E

$$\text{Efficiency}(\eta) = \frac{\text{Input Power}}{\text{Output Power}} \times 100$$

Taking the Source Voltage 12V:

Input Power : 4.2W

Output Power: 3.8136W

Efficiency: 90.80%

For 12V Input:

$$\text{Ripple current } (\Delta I_L) = \Delta V \times (2\pi \times f \times C)$$

Where: $f = 150 \text{ Hz}$, $C = 471 \mu\text{F}$ ($470 \mu\text{F} + 1 \mu\text{F}$)

$\Delta V = 800 \text{ mV}$ (max ripple)

$$\Delta I_L = 0.80 \times 2\pi \times 150 \times 471 \times 10^{-6}$$

$$\Delta I_L = 590 \text{ mA}$$

$$\text{Line Regulation} = \frac{[V_{\text{out_max}} - V_{\text{out_min}}]}{V_{\text{out_nominal}}} \times 100\%$$

$$\text{Line Regulation} = \frac{[(4.64 \text{ V} - 4.52 \text{ V})/5 \text{ V}]}{5 \text{ V}} \times 100\% = \frac{[0.12/5]}{5 \text{ V}} \times 100\% = 2.4\%$$

Voltage Regulation = $[(V_{out_high_input} - V_{out_low_input}) / V_{out_low_input}] \times 100\%$
Voltage Regulation = $[(4.64V - 4.58V) / 4.58V] \times 100\% = [0.06 / 4.58] \times 100\% = 1.31\%$

Results and Challenges

- Significant output voltage fluctuations
- High output ripple
- Thermal hotspots due to poor component placement
- EMI issues due to long trace paths

Lessons Learned

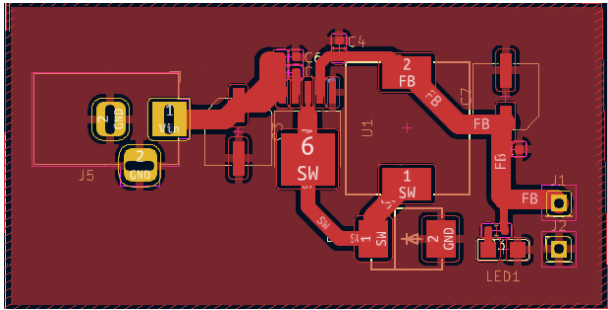
- Critical importance of proper component placement
- Need for calculated values rather than estimated ones
- Significance of proper grounding
- Importance of bypass capacitors

3.2 Second and Third Iteration

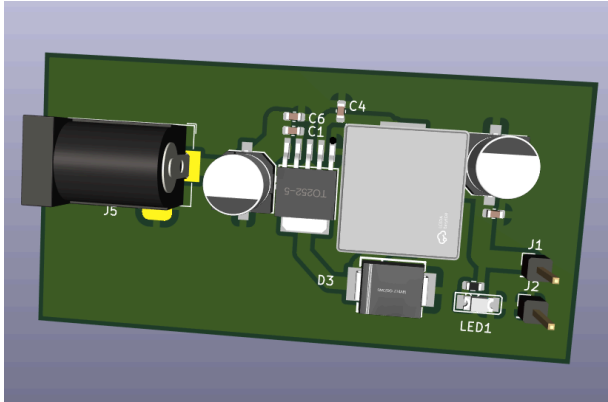
This iteration incorporated proper calculations and improved layout techniques:

Design Improvements

- Calculated component values based on:
 - Load requirements
 - Ripple specifications
 - Switching frequency considerations
- Board size reduced to 60mm × 30mm
- Added ceramic bypass capacitors
- Implemented proper grounding



PCB Layout



3D Model

Observations and Calculations:

Source Voltage(V)	Source current(A)	Average Voltage(V)	Range Fluctuation(mV)	Current Drawn(A)	Remarks (load)
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12	0.86	4.98	500-600	1.87	2A
12	0.41	5	360-500	0.96	1A

Efficiency(η)= $\frac{\text{Input Power}}{\text{Output Power}} \times 100 \%$

Taking the Source Voltage 12V:

Input Power : 10.32W

Output Power: 8.964W

Efficiency: 86.88%

For 2A load:

$\Delta V = 600\text{mV}$ (max ripple)
 $XC = \frac{1}{(2\pi \times f \times C)}$
Where, C = 470 μF + 1 μF
 $\Delta IL = \Delta V/(XC)$
 $\Delta IL = 442\text{mA}$

For 1A load:

$\Delta V = 500\text{mV}$ (max ripple)
 $\Delta IL = 368\text{mA}$

Component Updates

- Inductor upgraded to calculated 47 μH
- Added 1 μF ceramic bypass capacitor at input
- Added 100nF ceramic bypass capacitor at output
- Upgraded input capacitor to 330 $\mu\text{F}/25\text{V}$
- Upgraded output capacitor to 330 $\mu\text{F}/25\text{V}$

Results

- Improved voltage stability
- Reduced ripple voltage
- Better thermal distribution

3.3 Fourth Iteration

The final iteration focused on optimization and industrial-grade performance:

Design Optimization

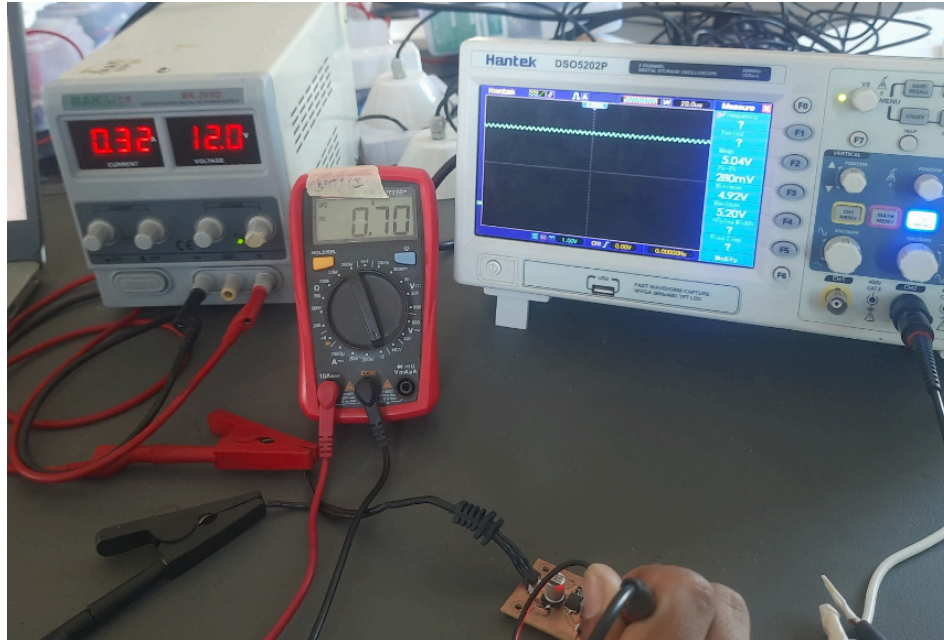
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3D Model

Observations and Calculations:

Source Voltage(V)	Source current(A)	Mean Voltage(V)	Range Fluctuation(mV)	Current Drawn(A)	Remarks (load)	Efficiency
12	0	5.04	160-200	0	No load	
12	0.34	5.04	240	0.74	audi qr	91.41176471
11.1	0.35	5.04	240	0.71	Audi qr	92.10810811
10.1	0.40	5.04	240	0.73	Audi qr	91.06930693
13	0.30	5.04	240-280	0.70	Audi qr	90.46153846
14	0.29	5.04	240-280	0.73	Audi qr	90.62068966
12	0.32	5.04	240-280	0.70	attendance	91.63636364
13	0.30	5.04	240-280	0.73	attendance	94.33846154
14	0.29	5.04	240-280	0.73	attendance	90.62068966
11.1	0.36	5.04	240	0.73	attendance	92.07207207
10	0.40	5.04	240	0.73	attendance	91.98
9	0.45	5.04	200-240	0.75	attendance	93.33333333
12	0.78	5.04	240-280	1.67	2A	89.92307692
13	0.74	5.04-5.08	240-280	1.63	2A	86.07484407
14	0.67	5.04	280-320	1.62	2A	87.04477612
11	0.85	5.04	240-280	1.66	2A	89.4802139
10	0.87	5.04	240-280	1.67	2A	96.74482759

- Input Voltage Range: 9-14V DC
- Output Mean Voltage : 5.04 V
- Switching Frequency: 148 kHz
- Output Voltage Ripple: 240 - 280mV
- Inductor ripple current: 419mA
- Efficiency: 90% (+- 2%)



Load - Attendance

Key Improvements

- Implemented proper ground planes
- Minimized trace lengths for critical paths
- Added test points for easy debugging
- Optimized component placement for heat dissipation
- Enhanced EMI protection through proper layout

Performance

- Stable output voltage ($\pm 0.3V$)
- Low ripple voltage (200mV- 300mV)
- Improved thermal performance
- Proper efficiency achieved
- EMI reduction through proper shielding and layout

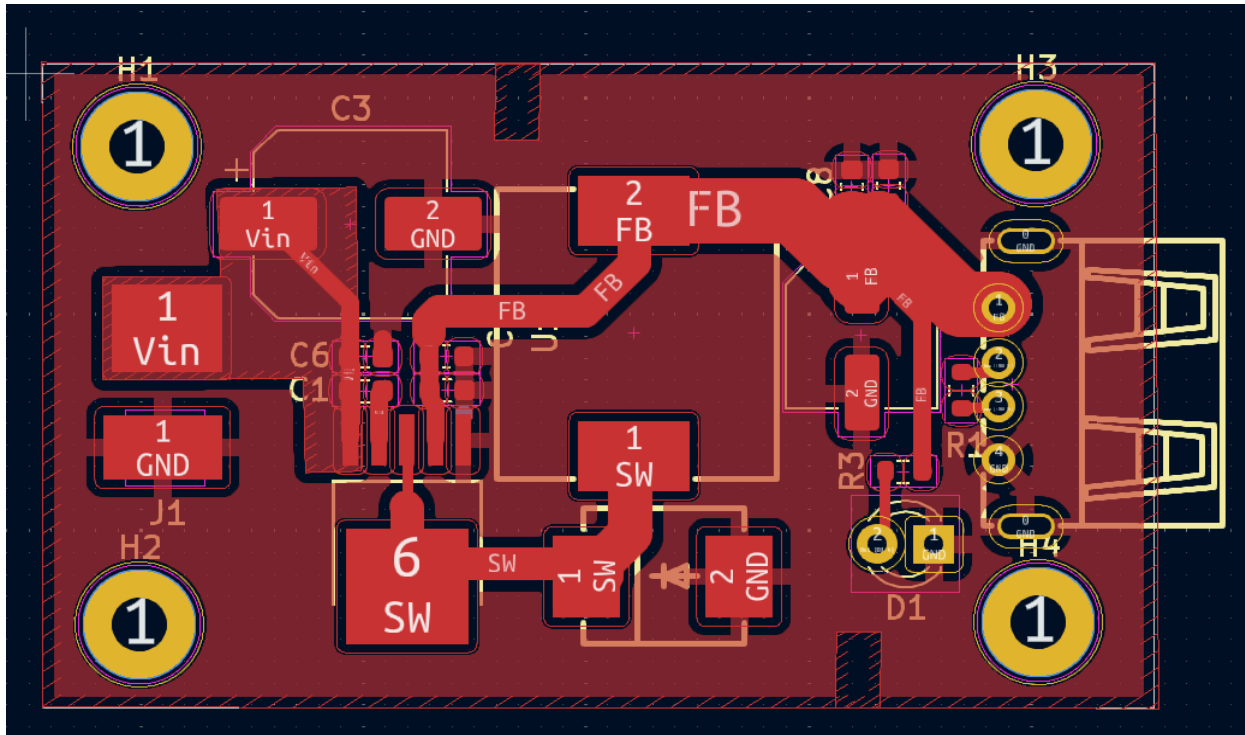
3.4 Fifth Iteration

Design Optimization

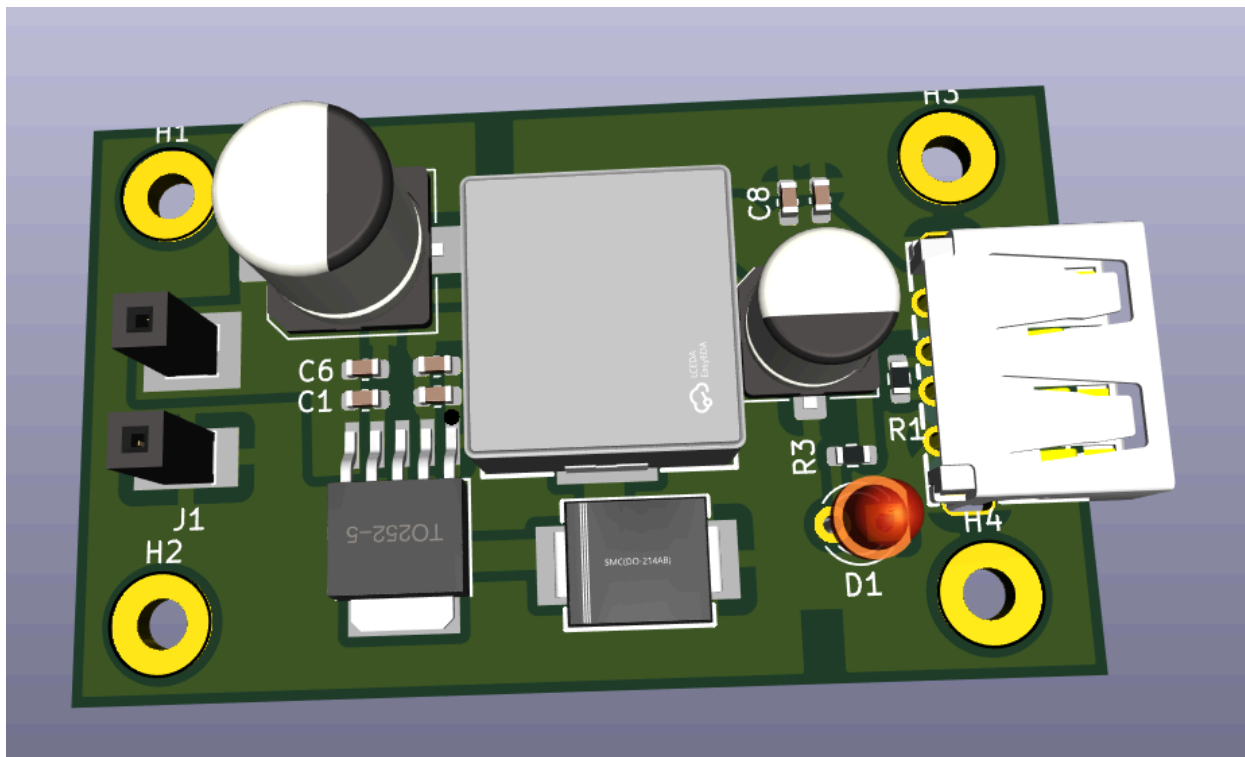
- Layout: 50mm $\times$ 30 mm
- Optimized component placement:
 - Input capacitors placed close to IC power pin
 - Minimized high-current trace lengths
 - Strategic bypass capacitor placement

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- The image displays four circuit diagrams for a USB 2.0 to RS485 converter, each within a rectangular frame.
- Input Terminal:** Shows the input signal path. It starts with a green line labeled V_{in} entering a red terminal block. The first terminal is connected to a capacitor $C3$ (330uF) to ground. The second terminal is connected to a capacitor $C6$ (1uF) to ground. The third terminal is connected to a capacitor $C1$ (1uF) to ground. The fourth terminal is connected to a green line labeled V_{in} and a red terminal block. The fifth terminal is connected to ground.
 - Buck Converter:** Shows the power regulation stage. It features a yellow block labeled $U2$ (XL2013E1). The input V_{in} is connected to pin 5. The feedback pin (FB) is connected to a network of capacitors $C4$ (1uF) and $C5$ (100nf) to ground, and a capacitor $C6$ (100nf) to the output. The output is connected to a switch (SW) and a red terminal block. The switch is connected to a green line labeled V_{out} and a red terminal block. The output is also connected to a red terminal block.
 - Inductor, Diode, Led:** Shows the output filter and indicator. It starts with a green line labeled V_{out} entering a red terminal block. The first terminal is connected to an inductor $L1$ (10uH) to ground. The second terminal is connected to a capacitor $C2$ (1uF) to ground. The third terminal is connected to a capacitor $C7$ (470uF) to ground. The fourth terminal is connected to a green line labeled V_{out} and a red terminal block. The fifth terminal is connected to ground. A red LED is connected between the output and ground.
 - Output Terminal:** Shows the output signal path. It starts with a green line labeled V_{out} entering a red terminal block. The first terminal is connected to a resistor $R1$ (200) to ground. The second terminal is connected to a green line labeled V_{out} and a red terminal block. The third terminal is connected to ground. The fourth terminal is connected to a green line labeled V_{out} and a red terminal block. The fifth terminal is connected to ground.

Schematic Diagram



PCB Layout



3D Diagram

Observations and Calculations:

Source Voltage(V)	Source current(A)	Mean Voltage(V)	Range Fluctuation(mV)	Current Drawn(A)	Remarks (load)	Efficiency
12	0	5.04	240	0	No load	
12	0.38	5.04	280-360	0.84	1A load	
13	0.34	5.04	280-360	0.80		
14	0.32	5.04	280-320	0.81		
12	0.82	5.04	600-900	1.75	2A load	
12	0.41	5.04	500-900	0.89	1A	

Component at 1A 40 to 50 degree Celsius
Component at 2A degree Celsius

3.5 Key Learning Outcomes

1. Component Selection:
 - o Proper calculation of passive components is crucial
 - o Quality of components directly impacts performance
 - o Bypass capacitors are essential for stability
2. PCB Layout:
 - o Short traces are critical for high-frequency circuits
 - o Ground plane design significantly affects performance
 - o Component placement affects thermal management
3. Design Methodology:
 - o Importance of systematic approach over trial and error
 - o Value of proper documentation and calculations
 - o Significance of iterative improvement process

4. Component Selection and Calculations

Inductor Selection

Inductor Value (L) = 47µH
 $\Delta I_L = (V_{in} - V_{out}) \times D / (L \times f)$
Where:
V_{in} = Input voltage = 12V

Vout = Output voltage = 5V
 D = Duty cycle = $V_{out}/V_{in} = 5/12 = 0.417$
 L = Inductance = 47 μ H
 f = Switching frequency = 148-153 kHz
 Ripple current = 419mA

Capacitor Selection

Input Capacitors

- Electrolytic Capacitor: 330 μ F/25V
- Ceramic Capacitor: 1 μ F
- Bypass Capacitor in across XL2013E1: 1 μ F

Output Capacitors

- Main Electrolytic: 470 μ F/10V
- Ceramic Bypass: 100nF and 1 μ F
- Capacitor across the feedback: 1 μ F and 100nF

5. Component List and Pricing

Components	Manufacture Part	Part Number(Lcsc)	Values	Number of component used	Price (1000 pieces)	Unit price	Rate
Inductor	HA127-470MT	C2961329	47uH	1	\$311	\$0.311	\$0.311
USB Type- A	A/F90 D10.6PBT6.5	C42434	-	1	\$48	\$0.048	\$0.048
Buck Dc to Dc	XL2013E1	C73339	-	1	\$240.30	\$0.2403	\$0.2403
Diode	B540C-13-F	C72264	-	1	\$57.90	\$0.0579	\$0.0579
Electrolytic Capacitors - SMD	CS1E331M-C RF10	C5162349	330uF-25v	1	\$37	\$0.037	\$0.037
Electrolytic Capacitors - SMD			470uF-10v	1	\$6.60	\$0.0066	\$0.0066
Ceramic Capacitor	CS1608Y5V1 05Z160NRB	C5189821	1nF	4	\$2.50	\$0.0025	\$0.01
Ceramic Capacitor	C0805B104K 500NT	C476766	100nF	2	\$2.10	\$0.0021	\$0.0042
Led				1		\$0.01	\$0.01

Resistor SMD			200 ohm	1		\$0.01	\$0.01
Resistor SMD			1000 ohm	1		\$0.01	\$0.01
Total				15			\$0.745

6. Conclusion

The buck converter design meets all specified requirements:

- Achieves 90% efficiency
- Maintains output voltage regulation within 0.5%
- Output ripple voltage with 200-320mV.
- Thermal performance within safe operating range
- Compact design suitable for industrial applications